Tonmya_Prescriptions

2026-04-18

Some simple models based on Tonmya sales

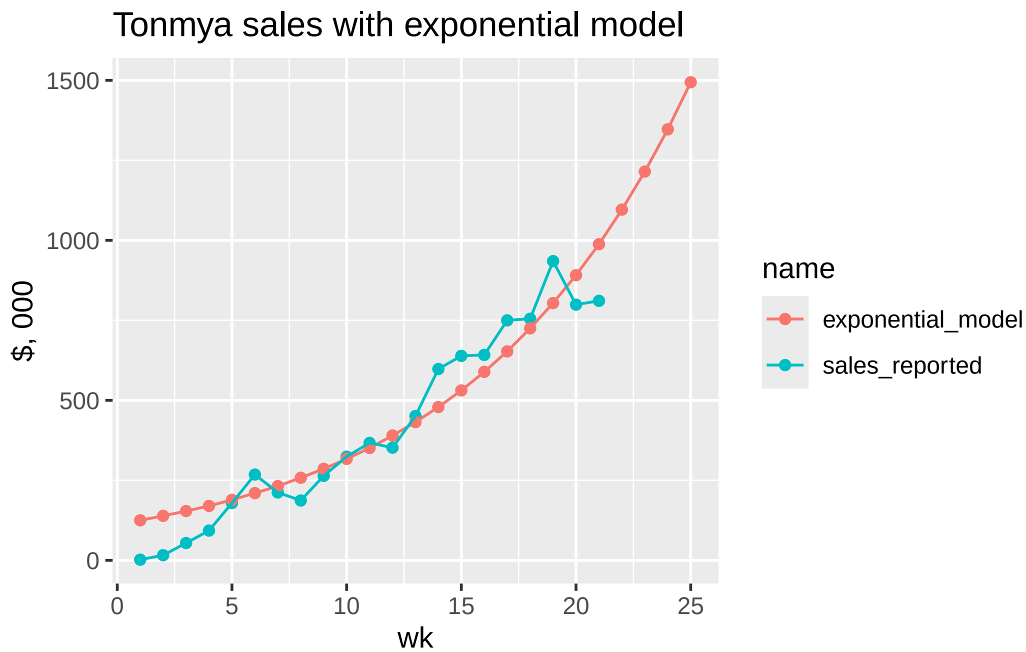
What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

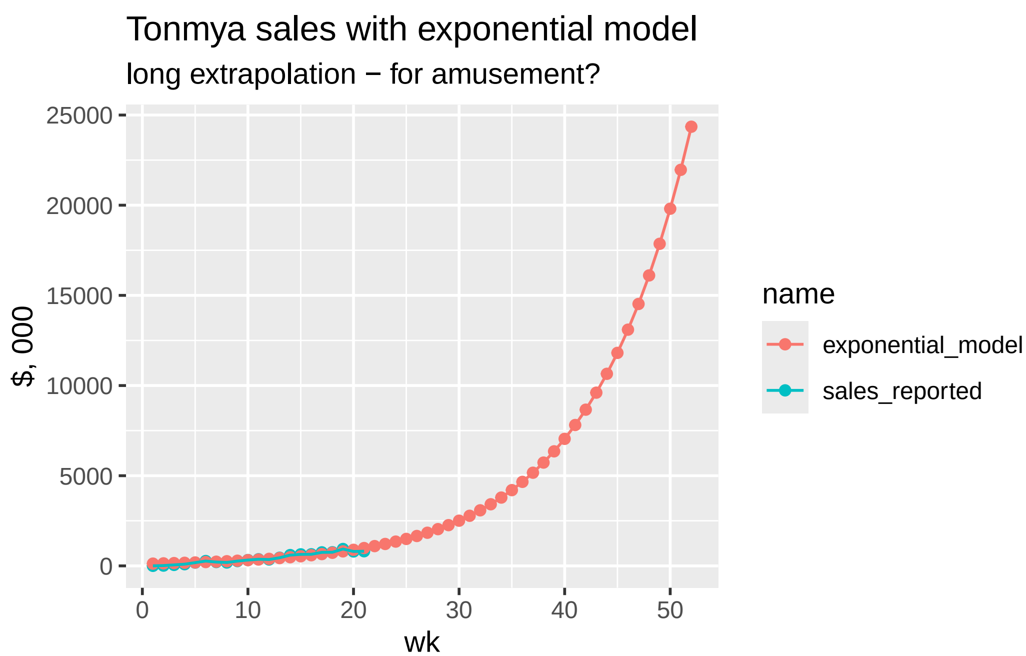
Sales Projection - at weekly growth rate - *exponential model*

```
Formula: sales ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 112.72828   18.62195   6.054 8.02e-06 ***
r   0.10891    0.01051  10.362 2.96e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 89.85 on 19 degrees of freedom

Number of iterations to convergence: 14
Achieved convergence tolerance: 2.197e-06
```

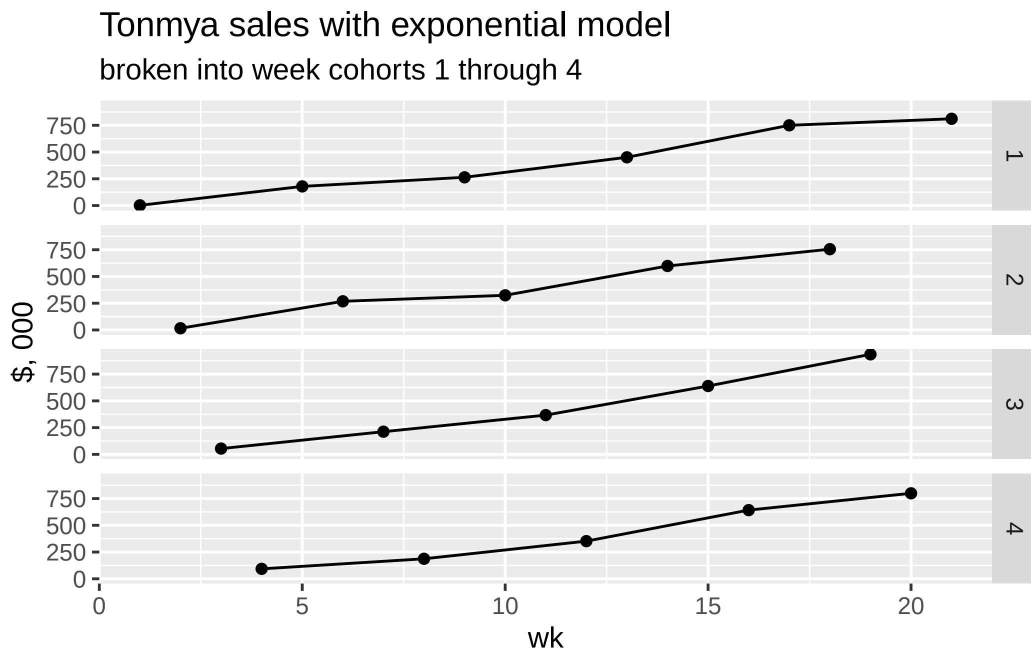
Exponential Model

wk	sales, 000	exp, 000	annual, M
1	$2	$125	$6.5
2	$16	$139	$7.2
3	$54	$154	$8.0
4	$93	$170	$8.9
5	$179	$189	$9.8
6	$268	$210	$10.9
7	$212	$232	$12.1
8	$187	$258	$13.4
9	$264	$286	$14.9
10	$324	$317	$16.5
11	$367	$351	$18.3
12	$352	$390	$20.3
13	$451	$432	$22.5
14	$598	$479	$24.9
15	$639	$531	$27.6
16	$642	$589	$30.6
17	$750	$653	$34.0
18	$755	$725	$37.7
19	$935	$804	$41.8
20	$799	$891	$46.3
21	$811	$988	$51.4
22	NA	$1,096	$57.0
23	NA	$1,215	$63.2
24	NA	$1,347	$70.1
25	NA	$1,494	$77.7





I think we can see some trending on a monthly basis. Reporting doesn't match a monthly basis, but we can see if four-week intervals are interesting.

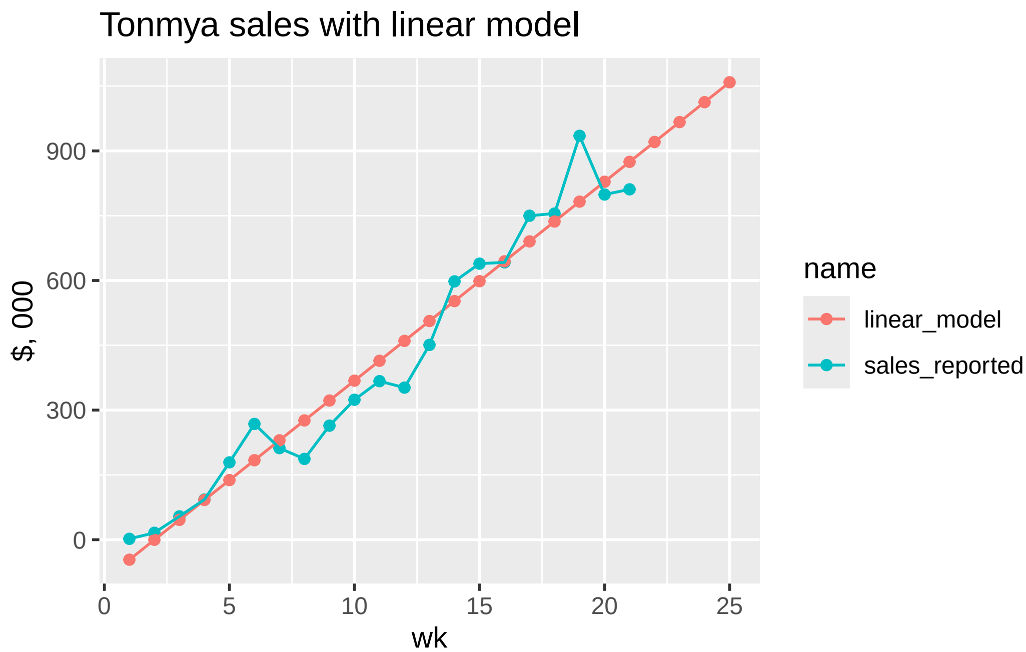


Sales Projection - at weekly rate - *linear model*

```
Call:
lm(formula = sales ~ wk, data = df_sales)

Residuals:
     Min       1Q   Median       3Q      Max
-108.240  -47.190    1.155   41.106  152.415

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -92.352     28.953   -3.19  0.00482 **
wk            46.049      2.306   19.97 3.27e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 63.98 on 19 degrees of freedom
Multiple R-squared:  0.9545,    Adjusted R-squared:  0.9521
F-statistic: 398.8 on 1 and 19 DF,  p-value: 3.27e-14
```

Linear Model

wk	sales, 000	exp, 000	annual, M
1	$2	-$46	$2.4
2	$16	-$0	$4.8
3	$54	$46	$7.2
4	$93	$92	$9.6
5	$179	$138	$12.0
6	$268	$184	$14.4
7	$212	$230	$16.8
8	$187	$276	$19.2
9	$264	$322	$21.6
10	$324	$368	$23.9
11	$367	$414	$26.3
12	$352	$460	$28.7
13	$451	$506	$31.1
14	$598	$552	$33.5
15	$639	$598	$35.9
16	$642	$644	$38.3
17	$750	$690	$40.7
18	$755	$737	$43.1
19	$935	$783	$45.5
20	$799	$829	$47.9
21	$811	$875	$50.3
22	NA	$921	$52.7
23	NA	$967	$55.1
24	NA	$1,013	$57.5
25	NA	$1,059	$59.9

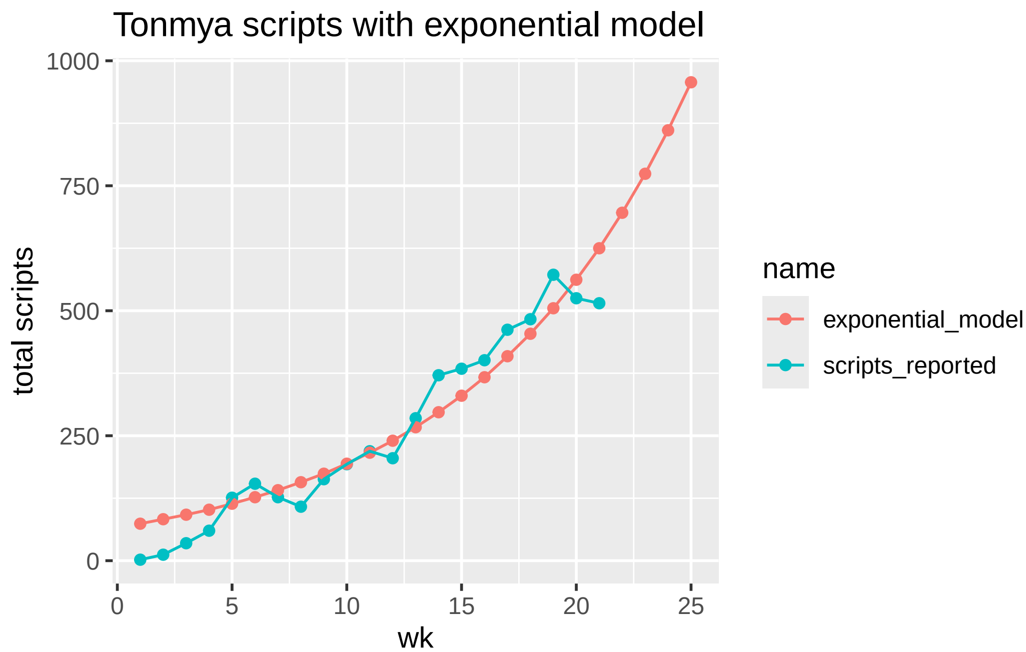


Scripts Projection - at weekly growth rate - *exponential model*

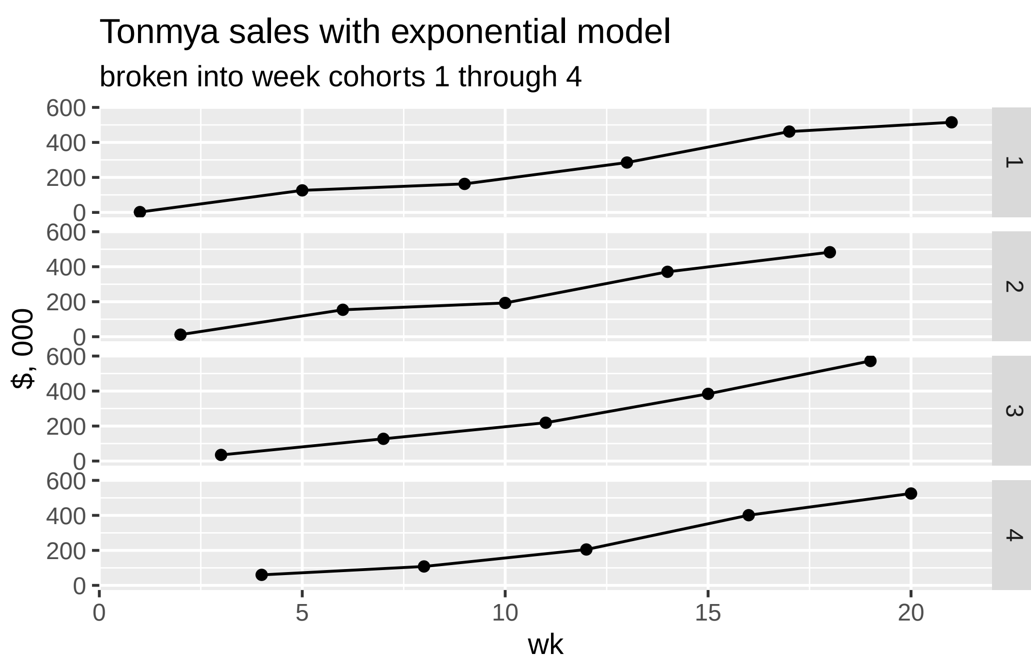
```
Formula: scripts ~ a * (1 + r)^wk

Parameters:
  Estimate Std. Error t value Pr(>|t|)
a 66.87705   10.48825   6.376 4.08e-06 ***
r  0.11233    0.00996  11.278 7.33e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 52.14 on 19 degrees of freedom

Number of iterations to convergence: 11
Achieved convergence tolerance: 7.34e-06
```

Scripts, Exponential Model

wk	total reported, 000	exp predicted, 000	annual, M
1	2	74	3.9
2	12	83	4.3
3	35	92	4.8
4	60	102	5.3
5	126	114	5.9
6	154	127	6.6
7	127	141	7.3
8	108	157	8.1
9	163	174	9.1
10	193	194	10.1
11	219	216	11.2
12	205	240	12.5
13	285	267	13.9
14	371	297	15.4
15	384	330	17.2
16	401	367	19.1
17	462	409	21.2
18	483	454	23.6
19	572	505	26.3
20	525	562	29.2
21	515	625	32.5
22	NA	696	36.2
23	NA	774	40.2
24	NA	861	44.8
25	NA	957	49.8

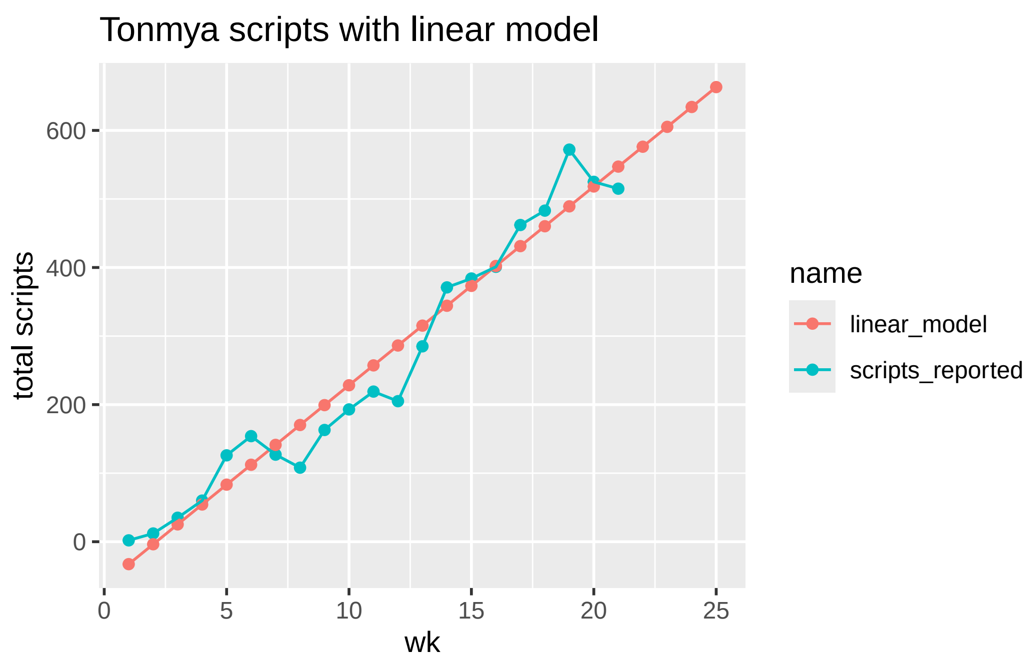


TRx Total Prescriptions by Weekly Cohorts



scripts Projection - at weekly rate - *linear model*

```
Call:
lm(formula = scripts ~ wk, data = df_scripts)

Residuals:
    Min      1Q  Median      3Q     Max 
-81.235 -32.212   6.785  26.770  82.783 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -61.733     18.277  -3.378  0.00316 ** 
wk            28.997      1.456  19.921 3.42e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.39 on 19 degrees of freedom
Multiple R-squared:  0.9543,	Adjusted R-squared:  0.9519 
F-statistic: 396.9 on 1 and 19 DF,  p-value: 3.422e-14
```



Scripts, Linear Model

wk	total reported, 000	linear predicted, 000	annual, M
1	2	-32.7	1.5
2	12	-3.7	3.0
3	35	25.3	4.5
4	60	54.3	6.0
5	126	83.3	7.5
6	154	112.3	9.0
7	127	141.2	10.6
8	108	170.2	12.1
9	163	199.2	13.6
10	193	228.2	15.1
11	219	257.2	16.6
12	205	286.2	18.1
13	285	315.2	19.6
14	371	344.2	21.1
15	384	373.2	22.6
16	401	402.2	24.1
17	462	431.2	25.6
18	483	460.2	27.1
19	572	489.2	28.6
20	525	518.2	30.2
21	515	547.2	31.7
22	NA	576.2	33.2
23	NA	605.2	34.7
24	NA	634.2	36.2
25	NA	663.2	37.7